THE CATEGORY OF COHERENT FORMAL MODULES IS ABELIAN: FOR X NOETHERIAN AND I A QUASI-COHERENT IDEAL SHEAF, COH(X, I) IS ABELIAN, RESTRICTION TO OPEN SUBSETS IS EXACT, AND EXACTNESS CAN BE CHECKED ON THE MEMBERS OF AN OPEN COVERING.

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6529-6541 of sources/stacks-coherent/pinned-original-source.tex; proof span is lines 6543-6610, the complete author proof (66 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Disclosure per the tightened ruling: the termwise cokernel claim carries '(details omitted)' and parts (2)-(3) are called immediate from the construction; the entire kernel construction and its Artin-Rees analysis - the load-bearing content - is written in the span.

Editorial note. Span computed from the \label marker through the enclosing \end{lemma}; precard_gate RESULT=OK (probe L9999).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-inverse-systems-abelian' also exists in stacks-spaces-more-morphisms — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. See why-field; every pointer named here occurs verbatim in the retained statement/proof text.

Source reference. Original lemma 00IN: sources/stacks-coherent/pinned-original-source.tex, source lines 6610–6610. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0316: sources/stacks-coherent/pinned-original-source.tex, source lines 6610–6610. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00IO: sources/stacks-coherent/pinned-original-source.tex, source lines 6610–6610. Cited by the retained statement or proof but

Date: stacks-project:coherent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 087W: sources/stacks-coherent/pinned-original-source.tex, source lines 6489–6495. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let X be a Noetherian scheme and let $\mathcal{I} \subset \mathcal{O}_X$ be a quasi-coherent sheaf of ideals.*

1. *The category $Coh(X, \mathcal{I})$ is abelian.*
2. *For $U \subset X$ open the restriction functor $Coh(X, \mathcal{I}) \to Coh(U, \mathcal{I}|_U)$ is exact.*
3. *Exactness in $Coh(X, \mathcal{I})$ may be checked by restricting to the members of an open covering of X.*

Proof. Let $\alpha = (\alpha_n) : (\mathcal{F}_n) \to (\mathcal{G}_n)$ be a morphism of $Coh(X, \mathcal{I})$. The cokernel of α is the inverse system $(\mathrm{Coker}(\alpha_n))$ (details omitted). To describe the kernel let

$$\mathcal{K}'_{l,m} = \mathrm{Im}(\mathrm{Ker}(\alpha_l) \to \mathcal{F}_m)$$

for $l \geq m$. We claim:

(a) the inverse system $(\mathcal{K}'_{l,m})_{l \geq m}$ is eventually constant, say with value $\mathcal{K}'_m$,
(b) the system $(\mathcal{K}'_m / \mathcal{I}^n \mathcal{K}'_m)_{m \geq n}$ is eventually constant, say with value $\mathcal{K}_n$,
(c) the system $(\mathcal{K}_n)$ forms an object of $Coh(X, \mathcal{I})$, and
(d) this object is the kernel of α.

To see (a), (b), and (c) we may work affine locally, say $X = \mathrm{Spec}(A)$ and $\mathcal{I}$ corresponds to the ideal $I \subset A$. By Lemma 087W α corresponds to a map $f : M \to N$ of finite $A^\wedge$-modules. Denote $K = \mathrm{Ker}(f)$. Note that $A^\wedge$ is a Noetherian ring (Algebra, Lemma 0316). Choose an integer $c \geq 0$ such that $K \cap I^n M \subset I^{n-c} K$ for $n \geq c$ (Algebra, Lemma 00IN) and which satisfies Algebra, Lemma 00IO for the map f and the ideal $I^\wedge = IA^\wedge$. Then $\mathcal{K}'_{l,m}$ corresponds to the A-module

$$K'_{l,m} = \frac{a^{-1}(I^l N) + I^m M}{I^m M} = \frac{K + I^{l-c} f^{-1}(I^c N) + I^m M}{I^m M} = \frac{K + I^m M}{I^m M}$$

where the last equality holds if $l \geq m + c$. So $\mathcal{K}'_m$ corresponds to the A-module $K / K \cap I^m M$ and $\mathcal{K}'_m / \mathcal{I}^n \mathcal{K}'_m$ corresponds to

$$\frac{K}{K \cap I^m M + I^n K} = \frac{K}{I^n K}$$

for $m \geq n + c$ by our choice of c above. Hence $\mathcal{K}_n$ corresponds to $K / I^n K$.

We prove (d). It is clear from the description on affines above that the composition $(\mathcal{K}_n) \to (\mathcal{F}_n) \to (\mathcal{G}_n)$ is zero. Let $\beta : (\mathcal{H}_n) \to (\mathcal{F}_n)$ be a morphism such that $\alpha \circ \beta = 0$. Then $\mathcal{H}_l \to \mathcal{F}_l$ maps into $\mathrm{Ker}(\alpha_l)$. Since $\mathcal{H}_m = \mathcal{H}_l / \mathcal{I}^m \mathcal{H}_l$ for $l \geq m$ we obtain a system of maps $\mathcal{H}_m \to \mathcal{K}'_{l,m}$. Thus a map $\mathcal{H}_m \to \mathcal{K}'_m$. Since $\mathcal{H}_n = \mathcal{H}_m / \mathcal{I}^n \mathcal{H}_m$ we obtain a system of maps $\mathcal{H}_n \to \mathcal{K}'_m / \mathcal{I}^n \mathcal{K}'_m$ and hence a map $\mathcal{H}_n \to \mathcal{K}_n$ as desired.

To finish the proof of (1) we still have to show that Coim = Im in $Coh(X, \mathcal{I})$. We have seen above that taking kernels and cokernels commutes, over affines, with the description of $Coh(X, \mathcal{I})$ as a category of modules. Since Im = Coim holds in the category of modules this gives Coim = Im in $Coh(X, \mathcal{I})$. Parts (2) and (3) of the lemma are immediate from our construction of kernels and cokernels. □